ELEC3104: Digital Signal Processing Lab

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Topic 1 – Time Domain Analysis (page 19)

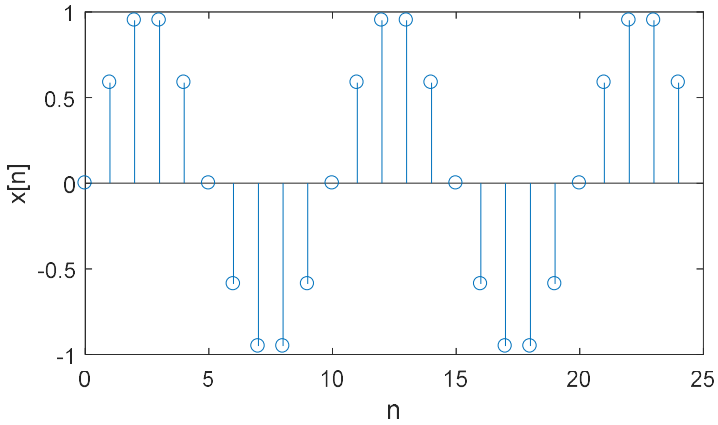
Exercise: Given a $x[n]$ by the equation:

$$x[n] = sin(\frac{2\pi n f_a}{f_s}),$$

where $f_a = 100Hz$, $f_s = 1\text{k}Hz$, $n = 0,1,2, \dots, 24$.

a. Write a MATLAB script to generate and plot x[n].

```
n=0:24;
fa=100;
fs=1000;
x=sin(2*pi*n*fa/fs);
figure
stem(n,x)
xlabel('n')
ylabel('x[n]')
```



b. Count the number of samples in one period of x[n] and verify it from the equation given above

It can be counted from the above figure that there are 10 samples in one period. Based on analytical equation, the number of samples in one period is $N=f_s/f_a=10$

Topic 1 – Frequency Domain Analysis (page 21)

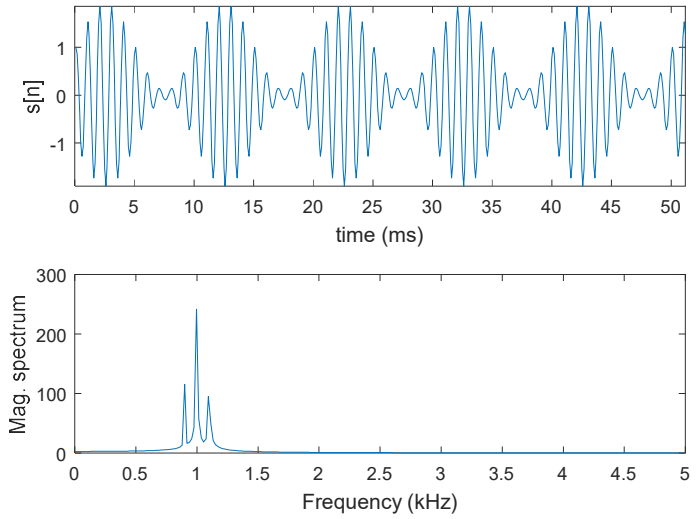
Exercise: An amplitude modulated signal s[n] is given by,

s[n]=[1+$msin(2\pi nf_m/f_s)$]cos($2\pi nf_c/f_s$), where m = 0.9, f_m=100Hz, f_c=1kHz, f_s=10kHz, n=0,1,2,…,511.

a. Write a MATLAB program to generate and plot s[n].

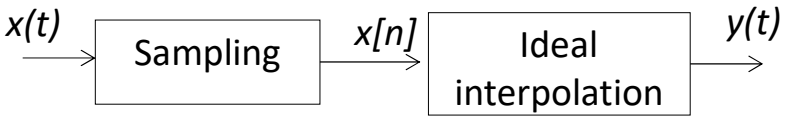
b. Plot the magnitude spectrum of s[n] in the frequency region [0, f_s/2] Hz and identify the frequencies present in the signal. -> the magnitude spectrum plot shows three frequencies present in the signal are around 0.9, 1, and 1.1 kHz

```
m = 0.9; fm=100; fc=10^3; fs=10^4;
n=0:511;
time_ms = 10^3*n/fs;
s=(1+m*sin(2*pi*n*fm/fs)).*cos(2*pi*n*fc/fs);
NFFT = 1024;%number of frequency point when compute fft
freqHz=fs*(0:NFFT-1)/NFFT;%Hz scale [0, fs]
freqHz=freqHz(1:NFFT/2); %Hz scale [0, fs/2]
S=fft(s,NFFT); %manitude spectrum in [0, fs] Hz
S=S(1:NFFT/2); %manitude spectrum in [0, fs/2] Hz
figure
subplot 211
plot(time_ms, s)
xlabel('time (ms)')
ylabel('Magnitude')
title('waveform of s[n]')
subplot 212
plot(freqHz*10^-3, abs(S));
ylabel('Magnitude')
xlabel('Frequency (kHz)')
title('Magnitude spectrum')
```



Topic 1 – Sampling Theorem and Aliasing (2), page 24

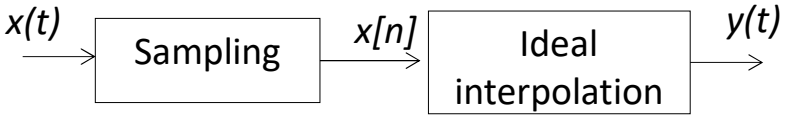
Exercise : An analogue signal *x(t) = 3cos(4000πt)+5sin(6000πt) +10cos(12000πt)* is sampled at a rate of f_s as the following figure to obtain the discrete-time signal x[n]



a. What is the minimum sampling frequency of f_s such that the sampling process does not cause an aliasing effect?
   f_1 = 2kHz, f_2=3kHz, f_3=6kHz -> f_{max}=6kHz
   $f_s \geq 2f_{max}$=12kHz, minimum sampling frequency is 12kHz
b. What is the analogue signal y(t) that can be reconstructed from x[n] if the ideal interpolation is used in the following cases:
   i. f_s=15kHz; 15kHz > $2f_{max}$ -> y(t) is same as x(t) because there is no aliasing
   ii. f_s=10kHz; -> f_1 , $f_2 < f_s/2$ hence no aliasing for these two frequencies;
   $f_3 > f_s/2$, this frequency component will be aliasing hence $f_3' = f_3 - f_s = 6-10 = -4$kHz
   y(t) = 3cos(4000πt)+5sin(6000πt) +10cos(-8000πt) = 3cos(4000πt)+5sin(6000πt) +10cos(8000πt)
   iii. f_s=7kHz (similar to question ii)
   iv. f_s=5kHz (similar to question ii)

Topic 1 – Sampling Theorem and Aliasing (2), page 24

Exercise : An analogue signal *x(t) = 3cos(4000πt)+5sin(6000πt) +10cos(12000πt)* is sampled at a rate of f_s as the following figure to obtain the discrete-time signal x[n]



c. Verify your answers in (b) by generating a 1000-sample x[n]. Then compute and plot the magnitude spectrum of x[n] versus Hz scale and estimate the frequency components of x[n] in the region [0, f_s/2]

```
f1=2000; f2=3000; f3=6000;
fs=15000;%can be changed to different values as in part (b)
n = 0:999;
NFFT = 1024;%number of frequency points when compute fft
x=3*sin(2*pi*n*f1/fs)+5*sin(2*pi*n*f2/fs)+10*sin(2*pi*n*f3/fs);
freqHz=fs*(0:NFFT-1)/NFFT;%Hz scale [0, fs]
freqHz=freqHz(1:NFFT/2); %Hz scale [0, fs/2]
X=fft(x,NFFT); %spectrum in frequency region [0, fs]
X=X(1:NFFT/2); %spectrum in frequency region [0, fs/2]
figure
plot(freqHz, abs(X));
ylabel('Magnitude')
xlabel('Frequency (Hz)')
title(['fs=' num2str(fs*10^-3) ' kHz'])
```

